DDR5 SDRAM
Complete Reference Document

Based on JEDEC Standard JESD79-5 (July 2020)

DDR5-3200 through DDR5-6400 · LPDDR5 Cross-reference · DFI 5.2 Mapping

Pinout & Addressing | Init & Reset | Mode Registers | Commands | Timing

Refresh & RFM | Training | ECC & CRC | ODT | Speed Bi

Compiled from JESD79-5 · 496 pages · Covers SDRAM architecture, electrical, timing, MR0–MR63+DFE, all commands, speed bins 3200–6400

Contents

1 DDR5 Architecture Overview

1.1 Key Architectural Changes vs DDR4

- **16n prefetch**: BL16 native (vs 8n in DDR4); optional BL8 (BC8) and BL32
- **Split channel**: one DIMM = two independent 32-bit sub-channels (A & B), each with own DQS
- **On-die ECC**: SEC (Single-Error Correction) mandatory; no separate ECC DRAM needed for x4/x8
- **Bank groups**: x4/x8: 8 BG × 4 banks = 32 banks (8 Gb) or 32 banks (16 Gb+); x16: 4 BG × 4 = 16 banks
- **Command bus**: 14-bit CA bus replaces traditional RAS/CAS/WE; 2-cycle command encoding
- **New commands**: MPC, VrefCA, VrefCS; RFMab/RFMsb; WR_PARTIAL; BL32
- **PMIC on DIMM**: Power management on-module; VDD = 1.1 V (vs 1.2 V DDR4)
- **CRC**: Mandatory write CRC; read CRC optional
- **DFE**: Decision Feedback Equalization in DRAM Rx
- **Refresh Management (RFM)**: Rowhammer mitigation built into spec

1.2 Signal Comparison: DDR4 vs DDR5

Signal	DDR4	DDR5
Command bus	RAS_n/CAS_n/WE_n/BA/A[13:0]	CA[13:0] + CS_n (2-cycle)
Data width	x4/x8/x16	x4/x8/x16 (per sub-channel x4/x8)
Prefetch	8n	16n (BL16 default)
Clock	CK_t/CK_c	CK_t/CK_c
DQS	DQS_t/DQS_c per byte	DQS_t/DQS_c per byte
Alert	ALERT_n	ALERT_n
ZQ	ZQ pin, ZQCL/ZQCS via MRS	ZQ pin, ZQCAL via MPC
Power	VDD=1.2V, VDDQ=1.2V, VPP=2.5V	VDD=1.1V, VDDQ=1.1V, VPP=1.8V
ODT	Dynamic ODT via MRS	Dynamic ODT via MR + CA_ODT strap
Reset	RESET_n	RESET_n

2 Package, Pinout & Addressing

2.1 Package Specifications

- **Package**: MO-210 BGA, ball pitch 0.8 mm × 0.8 mm
- x4/x8: 13 electrical rows, 6 electrical columns (2 sets of 3)
- 3 depopulated columns between electrical columns

2.2 Pinout Description

Pin	Type	Description
CK_t, CK_c	Input	Differential clock. All inputs sampled on CK_t rising edge
CA[13:0]	Input	14-bit command/address bus. Latched on both CK edges for 2-cycle cmds
CS_n	Input	Chip Select. Active LOW. Selects device for command reception
DQ[3:0/7:0/15:0]	I/O	Data bus. Width depends on device configuration
DQS_t, DQS_c	I/O	Differential data strobe. Source-synchronous. 1 per byte
DM_n/TDQS	I/O	Data mask (write) / Termination DQS (x8). MR-controlled
RESET_n	Input	Async reset. Active LOW. Initializes all internal logic
ALERT_n	Output	Open-drain alert: CRC error, CA parity error, temp alert
ZQ	Ref	External ZQ calibration resistor (240 Ω ±1%)
VDD	PWR	Core supply 1.1 V nom
VDDQ	PWR	I/O supply 1.1 V nom
VPP	PWR	Activating power supply 1.8 V
VSS/VSSQ	GND	Core/IO ground
LBDQ, LBDQS	I/O	Loopback data/strobe (optional)

2.3 Addressing Tables

2.3.1 8 Gb Devices

Config	BG	BA	BG/Banks	Row	Col	Page
8 Gb x4	BG0–2	BA0–1	8/2/16	R0–R16	C0–C10	1 KB
8 Gb x8	BG0–2	BA0–1	8/2/16	R0–R16	C0–C9	1 KB
8 Gb x16	BG0–1	BA0–1	4/2/8	R0–R16	C0–C9	2 KB

2.3.2 16 Gb Devices (and higher: 32 Gb, 64 Gb)

Config	BG	BA	BG/Banks/Total	Row	Col	Page
16 Gb x4	BG0–2	BA0–1	8/4/32	R0–R16	C0–C10	1 KB
16 Gb x8	BG0–2	BA0–1	8/4/32	R0–R16	C0–C9	1 KB
16 Gb x16	BG0–1	BA0–1	4/4/16	R0–R16	C0–C9	2 KB
32 Gb x4	BG0–2	BA0–1	8/4/32	R0–R16	C0–C10	1 KB
32 Gb x8	BG0–2	BA0–1	8/4/32	R0–R16	C0–C9	1 KB
32 Gb x16	BG0–1	BA0–1	4/4/16	R0–R16	C0–C9	2 KB
64 Gb x4	BG0–2	BA0–1	8/4/32	R0–R17	C0–C10	1 KB
64 Gb x8	BG0–2	BA0–1	8/4/32	R0–R17	C0–C9	1 KB
64 Gb x16	BG0–1	BA0–1	4/4/16	R0–R17	C0–C9	2 KB

3DS (3D Stacked): CID[3:0] selects die in stack. Max stack: 8H (x4/x8) or 4H (x16).

3 Reset and Initialization

3.1 Power-Up Initialization Sequence

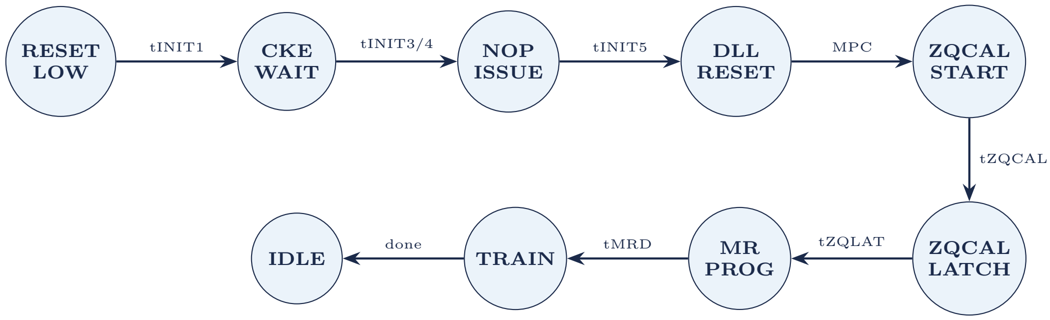
Steps (ref: JESD79-5 §3.3.1):

1. Apply VPP before or simultaneously with VDD/VDDQ
2. RESET_n held LOW for ≥ tINIT1 (200 µs) after voltage ramp completes
3. CS_n pulled LOW at least tINIT2 (10 ns) before RESET_n HIGH
4. After RESET_n HIGH: CS_n stays LOW for ≥ tINIT3 (4 ms)
5. DRAM registers EXIT on CS_n: wait tINIT4 (2 µs minimum)
6. Issue ≥ 3 NOP after CS_n HIGH (tINIT5)
7. Stable clock required: tCKSRX before any command
8. Issue **MPC DLL Reset**
9. Issue **MPC ZQCAL Start** → wait tZQCAL
10. Issue **MPC ZQCAL Latch** → wait tZQLAT
11. Program MRs (MR0, MR2, MR3, MR4, MR5, MR6, MR8, MR32–35, MR34–35)
12. Perform training (WL, read gate, read/write DQ, CA, CS, VREF)

3.2 Initialization Timing Parameters

Parameter	Symbol	Min	Max	Unit
Max voltage ramp time	tINIT0	–	20	ms
RESET_n LOW after voltage ramp	tINIT1	200	–	µs
CS_n LOW before RESET_n HIGH	tINIT2	10	–	ns
CS_n LOW after RESET_n HIGH	tINIT3	4	–	ms
DRAM EXIT registration	tINIT4	2	–	µs
NOP cycles after CS HIGH	tINIT5	3	–	nCK
Exit Reset to first config cmd	tXPR	tXS	–	–
Stable clock before command	tCKSRX	See SR tables	–	–
MR write period	tMRW	max(5 ns, 8 nCK)	–	nCK
MR read period	tMRR	max(14 ns, 16 nCK)	–	nCK
MR set delay	tMRD	max(14 ns, 16 nCK)	–	nCK

3.3 Init FSM State Diagram



3.4 MR Default Settings at Power-Up

Item	MR	Default	Description
Burst Length	MR0 OP[1:0]	00B	BL16
Read Latency	MR0 OP[6:2]	00010B	RL=26 @ DDR5-3200
Write Recovery	MR6 OP[3:0]	0000B	tWR=48 nCK @ 3200 or 30 ns
tRTP	MR6 OP[7:4]	0000B	12 nCK @ 3200 or 7.5 ns
VrefDQ	MR10	0010 1101B	75% VDDQ
VrefCA	MR11	0010 1101B	75% VDDQ
VrefCS	MR12	0010 1101B	75% VDDQ
ECS threshold	MR15	011B	256 errors
PPR	MR23 OP[1:0]	00B	hPPR & sPPR disabled
CK ODT	MR32 OP[2:0]	strap	Group A: RTT_OFF; Group B: 40 Ω
CS ODT	MR32 OP[5:3]	strap	Group A: RTT_OFF; Group B: 40 Ω
CA ODT	MR33 OP[2:0]	strap	Group A: RTT_OFF; Group B: 80 Ω
RTT_WR	MR34 OP[5:3]	001B	240 Ω
RTT_NOM_WR	MR35 OP[2:0]	011B	80 Ω

4 Mode Registers

4.1 MR Addressing Scheme

DDR5 uses LPDDR-style MR protocol: 8-bit **MRA** (Mode Register Address) + 8-bit **OP** (operand payload). Commands: **MRW** (write) and **MRR** (read), encoded on CA bus. Up to 256 addressable byte-wide registers (MR0–MR255).

4.2 MR0 — Burst Length / Read Latency

Bits	R/W	Field	Encoding
OP[1:0]	R/W	Burst Length	00B: BL16 (default) | 01B: BC8 (OTF) | 10B: BC8 fixed | 11B: BL32
OP[6:2]	R/W	CAS Latency (RL)	00000B: RFU | 00001B: CL22 | 00010B: CL26 | . . . see Table 25
OP[7]	RFU	–	Reserved

CL encoding (OP[6:2]): 00010B=26, 00011B=28, 00100B=30, 00101B=32, 00110B=34, 00111B=36, 01000B=38, 01001B=40, 01010B=42, 01011B=44, 01100B=46, 01101B=48, 01110B=50, 01111B=52, 10000B=54, 10001B=56, 10010B=58, 10011B=60, 10100B=62, 10101B=CL22 (bin A only).

4.3 MR1 — Drive Impedance / ODT

Bits	R/W	Field	Encoding
OP[2:0]	R/W	Pull-Down Drive Impedance	000B: RFU | 001B: 34 Ω (RZQ/7) | 010B: 40 Ω (RZQ/6) | 011B: 48 Ω (RZQ/5) | 100B: 60 Ω | 101B: 80 Ω | 110–111B: RFU
OP[5:3]	R/W	Pull-Up Drive Impedance	Same encoding as OP[2:0]
OP[6]	R/W	CS Assertion Duration	0B: 1-cycle | 1B: 2-cycle
OP[7]	R/W	Internal Write Timing	0B: Disabled (default) | 1B: Enabled

4.4 MR2 — Functional Modes

Bits	R/W	Field	Encoding
OP[1:0]	R/W	Max Power Saving Mode (MPSM)	00B: disable | 01B: MPSM | others: RFU
OP[2]	R/W	Device 15 MPSM	0B: all devices | 1B: device 15 only
OP[3]	R/W	2N Mode	0B: 1N | 1B: 2N
OP[4]	R/W	Read Preamble Training	0B: disable | 1B: enable
OP[5]	R/W	Write Leveling Training	0B: disable | 1B: enable
OP[6]	R/W	Reserved	–
OP[7]	R/W	Internal Write Timing	0B: disable | 1B: enable

4.5 MR3 — DQS Write Leveling Alignment

Bits	R/W	Field	Encoding
OP[3:0]	R/W	WL Cycle Alignment Lower Byte	0000B: 0 tCK (def) | 0001B: −1 tCK | ...| 0110B: −6 tCK (opt up to −15)
OP[7:4]	R/W	WL Cycle Alignment Upper Byte (x16)	Same encoding; x16 only

4.6 MR4 — Refresh Settings

Bits	R/W	Field	Encoding
OP[2:0]	R (SR)	Refresh Rate	001B: 1× (<80°C) | 010B: 1× (80–85°C) | 011B: 2× (85–90°C) | 100B: 2× (90–95°C) | 101B: 2× (>95°C)
OP[3]	SR/W	Refresh Interval Rate Indicator	0B: not implemented | 1B: implemented
OP[4]	R/W	Refresh tRFC Mode	0B: tRFC normal | 1B: tRFC Fine Granularity (FGR)
OP[6:5]	RFU	–	–
OP[7]	R	TUF (Temperature Update Flag)	0B: no change | 1B: change since last MR4 read

4.7 MR5 — ODT / DM

Bits	R/W	Field	Encoding
OP[2:0]	R/W	RTT_PARK	000B: off | 001B: 240 Ω | 010B: 120 Ω | 011B: 80 Ω | 100B: 60 Ω | 101B: 48 Ω | 110B: 40 Ω | 111B: 34 Ω
OP[5]	R/W	DM Enable	0B: disable | 1B: enable
OP[6]	R/W	TDQS Enable (x8 only)	0B: disable | 1B: enable
OP[7]	R/W	Data Output Disable	0B: normal | 1B: tri-state DQ/DQS

4.8 MR6 — tWR / tRTP

Bits	R/W	Field	Encoding
OP[3:0]	R/W	tWR	0000B: 48 nCK/30 ns | 0001B: 56 nCK/35 ns | 0010B: 64 nCK/40 ns | 0011B: 72 nCK/45 ns | 0100B: 80 nCK/50 ns | 0101B: 96 nCK/60 ns | 0110B: 112 nCK | 0111B: 128 nCK | 1000B: 144 nCK | 1001B: 160 nCK | 1010B: 192 nCK | 1011B: 224 nCK | 1100B: 256 nCK
OP[7:4]	R/W	tRTP	0000B: 12 nCK/7.5 ns | 0001B: 16 nCK/10 ns | 0010B: 20 nCK/12.5 ns | others: see Table

4.9 MR8 — Burst Length / Training / CRC

Bits	R/W	Field	Encoding
OP[1:0]	R/W	Burst Length	00B: BL16 | 01B: BC8 OTF | 10B: BC8 fixed | 11B: BL32
OP[2]	R/W	Read Preamble	0B: 2 tCK | 1B: 4 tCK
OP[3]	R/W	Write Preamble	0B: 2 tCK | 1B: 4 tCK
OP[4]	R/W	Read Postamble	0B: 0.5 tCK | 1B: 1.5 tCK
OP[5]	R/W	Write Postamble	0B: 0.5 tCK | 1B: 1.5 tCK
OP[6]	R/W	Write CRC Enable	0B: disable | 1B: enable
OP[7]	R/W	Read CRC Enable	0B: disable | 1B: enable

4.10 MR10, MR11, MR12 — VREF Settings

MR	Bits	Function
MR10	OP[7:0]	VrefDQ global calibration value. Range: 35.0–97.5% VDDQ in 0.5% steps. Default: 0x2D = 75%
MR11	OP[7:0]	VrefCA global calibration value. 75% default
MR12	OP[7:0]	VrefCS global calibration value. 75% default

Per-pin VrefDQ trims available in MR118, MR126, MR134, ... MR254 (OP[7:4]), ±3 steps max. Combined global + per-pin must stay within 35.0–97.5%.

4.11 MR13 — RFU (reserved)

4.12 MR14 — ECS (ECC Scrub) Control

- OP[1:0]: ECS mode – 00B: disabled | 01B: ECS during self-refresh | 10B: ECS during normal operation
- OP[4:2]: ECS interval (tECSint)
- OP[7:5]: Transparency mode selector

4.13 MR15 — ECS Error Threshold

`OP[2:0]`: ECS Error Threshold Count (ETC): 000B=1, 001B=4, 010B=16, 011B=256 (default), 100B=1024, 101B=4096, 110B=16384, 111B=65536.

4.14 MR16–MR21 — ECC Error Logging

- MR16–MR19: Row address of max-error row (R[17:0] across 4 bytes)
- MR20: Number of rows/code words with errors per die
- MR21: RFU

4.15 MR23 — Post Package Repair (PPR)

`OP[1:0]`: 00B = hPPR/sPPR disabled (default) | 01B = hPPR | 10B = sPPR | 11B = RFU

4.16 MR24 — PPR Guard Key

Guard key encoding prevents accidental PPR: specific OP values required before PPR activation.

4.17 MR32 — CK/CS ODT Settings

Bits	R/W	Field	Encoding
OP[2:0]	R/W	CK ODT	000B: RTT_OFF | 001B: 240 Ω | 010B: 120 Ω | 011B: 80 Ω | 100B: 60 Ω | 101B: 48 Ω | 110B: 40 Ω | 111B: 34 Ω
OP[5:3]	R/W	CS ODT	Same encoding
OP[7:6]	RFU	–	–

4.18 MR33–MR35 — CA/DQS/RTT Settings

MR	Bits	Function
MR33	OP[2:0]	CA ODT (Group A: RTT_OFF, Group B: 80 Ω default)
MR33	OP[5:3]	DQS_RTT_PARK (000B: RTT_OFF default)
MR34	OP[2:0]	RTT_PARK (000B: RTT_OFF default)
MR34	OP[5:3]	RTT_WR (001B: 240 Ω default)
MR35	OP[2:0]	RTT_NOM_WR (011B: 80 Ω default)
MR35	OP[5:3]	RTT_NOM_RD (011B: 80 Ω default)

RTT encoding (all RTT fields): 000B=RTT_OFF, 001B=240 Ω, 010B=120 Ω, 011B=80 Ω, 100B=60 Ω, 101B=48 Ω, 110B=40 Ω, 111B=34 Ω

4.19 MR58 — Refresh Management (RFM)

Bits	R/W	Encoding
OP[0]	R	0B: RFM not required | 1B: RFM required
OP[4:1]	R	RAAIMT (Initial Management Threshold): values 0100B=32, 0101B=40, ...
OP[7:5]	R	RAAMMT (Maximum Management Threshold)

4.20 MR61 — Package Output Driver Test Mode

Test mode for DRAM package characterization. OP[4:0] selects test pin. Disabled = 00000B (default).

4.21 MR63 — DRAM Scratch Pad

Read/write scratch register. Used by host controller to read RCD Control Words. Any value valid; no function in DRAM itself.

4.22 MR70–MR255 — DFE, Per-Bit DCA & VrefDQ

Organized by DQ bit: DQL0 at MR128 (MA=0x80), DQL1 at MR136, ...DQU7 at MR248. Each group of 8 MRs covers: DFE tap coefficients, per-bit duty cycle adjuster (DCA), per-bit VrefDQ offset. MA[6:3] directly maps to DQ bit; MA[2:0] selects function within that DQ group.

5 Command Set

5.1 Command Encoding Overview

DDR5 uses a 2-cycle command protocol on CA[13:0] + CS_n:

- **Cycle 1**: CS_n LOW + CA[13:0] encodes command type + upper address/data
- **Cycle 2**: CS_n LOW or HIGH + CA[13:0] encodes lower address/data
- Commands decoded from CA[6:0] pattern on cycle 1
- **DES** (deselect): CS_n HIGH = no operation

5.2 Command Truth Table

Command	Abbr	CS_n C1	CA[13:0] C1 Key Bits / Function
Deselect	DES	H	Any — no command registered
Activate	ACT	L	CA[6]=L, CA[5]=L, CA[4:0]=bank/BG; C2: row address
Read	RD	L	CA[6]=L, CA[5]=H, CA[4]=L; AP, BG, BA, COL
Read w/ AP	RD/A	L	RD + CA[10]=H (auto-precharge)
Write	WR	L	CA[6]=H, CA[5]=L, CA[4]=L; AP, BG, BA, COL, WR_PARTIAL
Write w/ AP	WR/A	L	WR + CA[10]=H
Precharge (per bank)	PRE(pb)	L	CA[5:4]=01; CA[10]=L; BG, BA
Precharge All	PRE(ab)	L	CA[5:4]=01; CA[10]=H
Refresh All Banks	REFab	L	CA[6:4]=001
Refresh Same Bank	REFsb	L	CA[6:4]=010; BA[1:0]
Refresh Mgmt All	RFMab	L	CA encoding per Table
Refresh Mgmt Same Bank	RFMsb	L	BA-specific
Self Refresh Entry	SRE	L	CS pattern per Table
Self Refresh Exit	SRX	L	Requires tCKSRX stable clock before
Mode Register Write	MRW	L	CA[6:4]=011; C2: MRA[7:0] + OP[7:0]
Mode Register Read	MRR	L	CA[6:4]=100; C2: MRA[7:0]; DQ returns data
MPC (Multi-Purpose Cmd)	MPC	L	CA[13:7]=specific pattern; OP[7:0] = opcode
VrefCA Command	VrefCA	L	Dedicated CA encoding; OP sets VREF
VrefCS Command	VrefCS	L	Dedicated CS encoding; OP sets VREF
Write Pattern	WRP	L	MPC opcode 0000 0001

5.3 ACTIVATE Command

Opens a row in a bank for subsequent RD/WR.

- BG[2:0] (x4/x8) or BG[1:0] (x16): selects bank group
- BA[1:0]: selects bank within group
- R[17:0]: row address (C1+C2 combined)
- 3DS: CID[3:0] selects die in stack
- Bank-to-bank ACT timing: **tRRD_S** (different BG) or **tRRD_L** (same BG)
- Max 4 ACTs within tFAW window

5.4 READ / WRITE Commands

- CAS issued tRCD after ACT
- Read latency: **RL** = CL (set in MR0)
- Write latency: **WL** = CL−2 (CWL = CL−2, fixed relationship in DDR5)
- Data transfer: 16 UI per burst (BL16); DQ toggled on both DQS edges
- **AP** (CA10=H): auto-precharge after burst completes
- **WR_PARTIAL**: enables DM-masked partial writes; triggers internal RMW
- CAS-to-CAS: tCCD_L (same BG), tCCD_S (diff BG); tCCD_L_WR for WR-to-WR same BG

5.5 MPC Opcode Table (Key Opcodes)

OP[7:0]	MPC Command	Description
0000 0000	ZQCALstart	Initiates ZQ calibration start
0000 0001	ZQCALlatch	Latches ZQ calibration result
0000 0010	DLL Reset	Resets DLL
0000 0011	Write Pattern	Puts device in write pattern mode
0000 0100	RTP Reset	Resets read training pattern
0000 0101	RTP Start	Starts read training pattern output
0000 0110	RTP Stop	Stops read training pattern
0001 0000	CATraining	Enables CA training mode
0001 0001	CSTraining	Enables CS training mode
0001 0010	WLTraining	Enables write leveling
0001 0011	Preamble Training	Enables preamble training mode
0001 0100	Connectivity Test	Enters connectivity test (CT) mode
0001 0101	DQS Oscillator Start	Starts DQS interval oscillator
0001 0110	DQS Oscillator Stop	Stops DQS interval oscillator

6 Timing Parameters

6.1 Key Timing Definitions

Symbol	Name	Description
tCK(avg)	Clock period	Average clock period
tAA	CAS latency	CMD to first data. = CL × tCK
tRCD	RAS to CAS	ACT to RD/WR delay
tRP	Row Precharge	PRE to ACT same bank
tRAS	Row Active	ACT to PRE minimum
tRC	Row Cycle	ACT to ACT same bank = tRAS + tRP
tRFC	Refresh Cycle	REF to any cmd, same rank. Density-dependent
tRFCpb	REF per-bank	REFsb to next cmd same bank
tREFI	Refresh interval	Time between REF commands (7.8 μs @ <85°C)
tFAW	4-ACT window	Max 4 ACT cmds within this window
tCCD_L	CAS-to-CAS L	Same bank group RD-to-RD or WR-to-WR
tCCD_S	CAS-to-CAS S	Different BG RD-to-RD
tCCD_L_WR	WR-to-WR L	Same BG write-to-write
tCCD_L_WR2	WR-to-WR2	Same BG WR-to-WR, 2nd not RMW
tWTR_L	WR-to-RD L	Write end to RD cmd, same BG
tWTR_S	WR-to-RD S	Write end to RD cmd, diff BG
tRTW	RD-to-WR	$= t_{CL} + t_{BL/2} + 2 - t_{CWL}$
tWR	Write Recovery	WR cmd to PRE
tRTP	Read-to-Pre	RD cmd to PRE
tRRD_L	ACT-to-ACT L	Same BG
tRRD_S	ACT-to-ACT S	Different BG
tMRD	MR delay	MRW to any non-MR cmd
tMOD	MR operation	MRW to normal operation
tZQCAL	ZQ cal time	MPC ZQCALstart to ZQCALlatch
tDLLK	DLL lock	DLL Reset to ready

6.2 Timing Parameters by Speed Grade (DDR5-3200 to 4000)

Parameter	Symbol	3200		3600		4000		Unit
		min	max	min	max	min	max	
Clock period	tCK(avg)	0.625	<0.681	0.555	<0.625	0.500	<0.555	ns
CAS-to-CAS (same BG)	tCCD_L	max(8 nCK, 5 ns)						nCK
WR-to-WR (same BG)	tCCD_L_WR	max(32 nCK, 20 ns)						nCK
WR-to-WR2 (same BG)	tCCD_L_WR2	max(16 nCK, 10 ns)						nCK
CAS-to-CAS (diff BG)	tCCD_S	8 nCK						nCK
ACT-to-ACT (diff BG, 2KB)	tRRD_S(2K)	8 nCK						nCK
ACT-to-ACT (same BG)	tRRD_L	max(8 nCK, 5 ns)						nCK

6.3 Timing Parameters (DDR5-4400 to 5200)

Parameter	Symbol	4400		4800		5200		Unit
		min	max	min	max	min	max	
tCK(avg)	Clock period	0.454	<0.500	0.416	<0.454	0.384	<0.416	ns
tCCD_L	CAS same BG	max(8 nCK, 5 ns)						nCK
tCCD_S	CAS diff BG	8 nCK						nCK
tRRD_S	ACT diff BG	8 nCK						nCK
tRRD_L	ACT same BG	max(8 nCK, 5 ns)						nCK

6.4 Timing Parameters (DDR5-5600 to 6400)

Parameter	Symbol	5600		6000		6400		Unit
		min	max	min	max	min	max	
tCK(avg)	Clock period	0.357	<0.384	0.333	<0.357	0.312	<0.333	ns
tCCD_L	CAS same BG	max(8 nCK, 5 ns)						nCK
tCCD_L_WR	WR same BG	max(32 nCK, 20 ns)						nCK
tCCD_L_WR2	WR2 same BG	max(16 nCK, 10 ns)						nCK
tCCD_S	CAS diff BG	8 nCK						nCK
tRRD_S(2K)	ACT diff BG 2K	8 nCK						nCK
tRRD_L(2K)	ACT same BG 2K	max(8 nCK, 5 ns)						nCK
tRRD_L(1K)	ACT same BG 1K	max(8 nCK, 5 ns)						nCK

6.5 tRFC by Device Density

Density	tRFC1 (ns) min	tRFC2 (FGR, ns) min
8 Gb	195	130
16 Gb	295	160
32 Gb	410	220
64 Gb	550	300

6.6 tREFI Parameters

Mode	tREFI
1× refresh, <85°C	7.8 µs
2× refresh, 85–95°C	3.9 µs
FGR 2× mode	3.9 µs
FGR 4× mode	1.95 µs

7 Speed Bins

7.1 Standard Speed Bins: 3200 – 6400

Speed Bin	CL-nRCD-nRP	CWL	tAA (ns)	tRCD (ns)	tRP (ns)	tRAS (ns)
DDR5-3200A	22-22-22	20	13.75	13.75	13.75	32.0
DDR5-3200B	26-26-26	24	16.25	16.25	16.25	32.0
DDR5-3200C	28-28-28	26	17.50	17.50	17.50	32.0
DDR5-3600A	26-26-26	24	14.44	14.44	14.44	32.0
DDR5-3600B	28-28-28	26	15.56	15.56	15.56	32.0
DDR5-4000A	28-28-28	26	14.00	14.00	14.00	32.0
DDR5-4000B	30-30-30	28	15.00	15.00	15.00	32.0
DDR5-4400A	30-30-30	28	13.64	13.64	13.64	32.0
DDR5-4400B	32-32-32	30	14.55	14.55	14.55	32.0
DDR5-4800A	32-32-32	30	13.33	13.33	13.33	32.0
DDR5-4800B	34-34-34	32	14.17	14.17	14.17	32.0
DDR5-5200A	36-36-36	34	13.85	13.85	13.85	32.0
DDR5-5200B	38-38-38	36	14.62	14.62	14.62	32.0
DDR5-5600A	38-38-38	36	13.57	13.57	13.57	32.0
DDR5-5600B	40-40-40	38	14.29	14.29	14.29	32.0
DDR5-6000A	40-40-40	38	13.33	13.33	13.33	32.0
DDR5-6000B	42-42-42	40	14.00	14.00	14.00	32.0
DDR5-6400A	42-42-42	40	13.13	13.13	13.13	32.0
DDR5-6400B	46-46-46	44	14.38	14.38	14.38	32.0

Note: CWL = CL − 2 for all DDR5 bins. tRC = tRAS + tRP (32 ns + tRP).

7.2 nCK Conversion Formula

$$t_{\text{param\_nCK}} = \left\lfloor \frac{t_{\text{param\_ps}} \times 1000 + 990}{t_{\text{CK\_ps}} \times 1000} \right\rfloor$$

Example: tWR = 30 ns @ DDR5-3200 (tCK = 625 ps): $\lfloor(30000 \times 1000 + 990)/(625 \times 1000)\rfloor = \lfloor 48990/1000 \rfloor =$ 48 nCK

8 Refresh & Refresh Management (RFM)

8.1 Refresh Modes

REFab (All-Bank): Refreshes all banks simultaneously. Uses tRFC1 recovery time.

REFsb (Same-Bank): Per-bank-group refresh. BA[1:0] selects bank in every BG. Uses tRFCpb recovery. Allows interleaved bank access during other banks' refresh.

FGR (Fine Granularity): MR4 OP[4]=1. Enables 2× or 4× refresh rates. Used at elevated temperatures or when spec requires tighter REF scheduling.

8.2 Same-Bank Refresh (REFsb) Operation

REFsb cycles through banks BA[1:0]: 00→01→10→11 in order. Controller tracks internal bank counter. 8 REFsb = 1 REFab credit. Switching from FGR to normal mode: all banks must have even REF count, or an extra REFab needed.

8.3 Refresh Management (RFM)

Rowhammer mitigation. Controller monitors **RAA (Rolling Accumulated ACT)** count per bank.

- Each ACT increments RAA counter for that bank by 1
- **RAAIMT** (MR58 OP[4:1]): vendor-set threshold; when RAA ≥ RAAIMT → issue RFM
- **RAAMMT** (MR58 OP[7:5]): maximum threshold; must not exceed
- **RFMab**: refresh management for all banks
- **RFMsb**: refresh management for specific bank
- RAA counter decremented per REF command (amount in MR: OP per spec)
- DRAM signals RFM required via **ALERT_n** or MR58 OP[0]=1

RAAIMT values (MR58 OP[4:1]): 0100B=32 (normal) / 16 (FGR) | 0101B=40/20 | 0110B=48/24 | 0111B=56/28 | 1000B=64/32 | 1001B=80/40 | 1010B=96/48 | ...

9 Training Sequences

9.1 Write Leveling

- Enable via MR2 OP[5]=1
- DRAM samples DQS relative to CK; returns feedback on DQ
- Controller sweeps DQS delay; finds optimal WL delay
- Result stored in **MR3** (WL Internal Cycle Alignment): 0 to −15 tCK offset
- Disable MR2 OP[5]=0 when done; MR3 setting retained
- tWLMRD: WL mode to first DQS edge; tWLDQSEN: WL mode enable to DQS enable

9.2 Read Training

Read Training Pattern (MPC RTP Start):

- DRAM outputs predefined pattern on DQ during RD commands
- Pattern types: Data0/Data1, LFSR0/LFSR1 (selected via MR30/31)
- Controller adjusts read capture timing and RD postamble until pattern matches

- tRTP_train: timing for read training pattern

Read Gate Training (Preamble Training, MPC Preamble Training):

- MR2 OP[4]=1: enables preamble training mode
- DRAM extends read preamble; controller finds DQS gate

9.3 CA Training

- MPC opcode 0001 0000: enter CA training mode
- Controller sweeps CA timing; DRAM samples and echoes result on DQ
- Output: sampled CA pattern on DQ (Table 313/314 per interface width)
- tCADT: CA training data valid after MPC
- Exit: MPC opcode to exit, followed by MRW to restore settings

9.4 CS Training

- MPC opcode 0001 0001: enter CS training mode
- Sweeps CS timing; result echoed on DQ
- Useful for multi-rank systems to set per-rank CS delays

9.5 VREF Training

VrefCA: VrefCA command sweeps CA input VREF. Range: internal spec per MR11. **VrefCS:** VrefCS command sweeps CS VREF. Range per MR12. **VrefDQ:** MRW to MR10 (global) + MR118/126/... (per-pin). Sweep: 35–97.5% VDDQ.

9.6 DQS Interval Oscillator

- MPC DQS Osc Start/Stop: measures DQS-to-DQ skew drift (due to ΔV, ΔT)
- Controller reads result via MRR; computes tDQS2DQ offset
- Used for periodic re-training without full sequence
- tDQSOSC: oscillation count readout timing

10 On-Die ECC & CRC

10.1 On-Die ECC (SEC)

- Mandatory for DDR5 x4/x8; transparent to host
- **SEC** (Single Error Correction): corrects 1 bit per code word
- **DED** detection: detects but cannot correct 2-bit errors
- Code word: per-byte (9 bits data + parity for x8), or per-nibble for x4
- No separate ECC DRAM required on DIMM for x4/x8

10.2 ECS (ECC Transparency and Error Scrub)

Controlled by MR14–MR21:

- Background scrub of memory array to correct accumulated errors
- Modes: disabled | scrub during self-refresh | scrub during normal operation
- tECSint: average periodic interval between scrub operations
- Error tracking: MR16–MR19 log row address of max-error row
- MR20: count of rows with errors; MR15: threshold for action

10.3 Write CRC

- Enabled via MR8 OP[6]=1
- 4-bit CRC appended to each BL16 burst per byte lane
- **Polynomial**: $G(x) = x^4 + x + 1$
- CRC bits occupy UI[16:19] (beyond BL16 data)

- Error: DRAM asserts **ALERT_n**; host retransmits
- **Auto-disable**: after N consecutive CRC errors, CRC auto-disabled (prevents lockout)
- **BC8 mode**: CRC mapped per Table 366 (bit position adjusted)

10.4 Read CRC (Optional)

- MR8 OP[7]=1 enables read CRC
- DRAM appends CRC bits to read burst
- Read CRC Latency Adder: additional RL cycles for CRC computation (Table 365)

11 On-Die Termination (ODT)

11.1 DQ ODT Modes

Mode	MR	Description
RTT_NOM_WR	MR35 OP[2:0]	Nominal termination during write operations
RTT_NOM_RD	MR35 OP[5:3]	Nominal termination during read operations
RTT_WR	MR34 OP[5:3]	Dynamic termination during write (240 Ω default)
RTT_PARK	MR34 OP[2:0] / MR5 OP[2:0]	Park termination when ODT not active

11.2 CA/CS/CK ODT

- **CA_ODT strap**: sets Group A (RTT_OFF) or Group B (80 Ω) at power-up
- Fine tuning via MR32 (CK/CS ODT) and MR33 (CA/DQS ODT)
- **ODTL** (ODT Latency): $ODTL_{on} = WL - 2$; $ODTL_{off} = WL + BL/2$
- Dynamic ODT: automatically switches RTT during write vs read
- tADC: ODT transition time constraint

11.3 RTT Supported Values

RTT encoding (applies to RTT_NOM, RTT_WR, RTT_PARK, CA/CK/CS ODT fields):

Code	RTT Value
000B	RTT_OFF (Hi-Z)
001B	240 Ω (RZQ/1)
010B	120 Ω (RZQ/2)
011B	80 Ω (RZQ/3)
100B	60 Ω (RZQ/4)
101B	48 Ω (RZQ/5)
110B	40 Ω (RZQ/6)
111B	34 Ω (RZQ/7)

12 Electrical & AC/DC Specifications

12.1 DC Operating Conditions

Parameter	Symbol	Min	Nom	Max	Unit
Core supply	VDD	1.060	1.100	1.170	V
I/O supply	VDDQ	1.060	1.100	1.170	V
Activating supply	VPP	1.700	1.800	1.900	V
Absolute max VDD	–	–	–	1.500	V
Absolute max VPP	–	–	–	2.500	V

12.2 Output Driver Impedance (DQ/DQS)

Mode	Min (Ω)	Max (Ω)	Note
RONPd @ VOL (0.1×VDDQ)	0.3 RZQ	1.1 RZQ	–
RONPd @ VOM (0.8×VDDQ)	0.4 RZQ	1.1 RZQ	–
RONPd @ VOH (0.95×VDDQ)	0.4 RZQ	1.25 RZQ	–

RZQ = 240 Ω ±1%. All driver codes referenced to RZQ pin value.

12.3 Operating Temperature

Parameter	Value
Commercial	0 to +85°C
Industrial	−40 to +95°C
Extended	−40 to +105°C (if specified)

12.4 BER Requirements

- Minimum BER: 10^{-17} for Rx/Tx timing and voltage tests (DDR5-3200 to 6400)
- Confidence levels: 70%–99.5% (Table 393)
- Applied DCD: 0.045 UI; Random Jitter RMS: 0.0075 UI (Rx DQS jitter sensitivity test)

13 Advanced Features

13.1 Decision Feedback Equalization (DFE)

- In-DRAM Rx DQ equalizer; compensates for ISI
- **Components**: summing amplifier + feedback path with N taps
- Tap coefficients: MR70–MR255 (per-DQ DFE registers)
- Gain adjustment range: vendor-specified min/max (Table 345)
- Tap coefficient range per Table 346
- Update settling time: tDFE = 80 ns min
- Configuration: global via MRW; per-pin offsets via DFE MRs

13.2 Duty Cycle Adjuster (DCA)

Adjusts internal DQS clock duty cycle per DQ/DQS:

- 2-phase internal clock: DCA adjusts 50/50 duty cycle
- 4-phase: QCLK, IBCLK, QBCLK adjustments independent (Figs 149–151)
- Range: −3 to +3 LSB codes per DQ (MR70–MR255 DCA fields)

13.3 2N Mode

- MR2 OP[3]=1: 2N mode; MR2 OP[3]=0: 1N mode (default)
- In 2N mode: CS_n and CA sampled over 2 clock cycles instead of 1
- Used for timing margin recovery at high frequencies or long traces
- CS/CA required behavior per Table 353

13.4 Post Package Repair (PPR)

hPPR (Hard PPR): Permanently remaps a failing row to a spare row. Guard key in MR24 required. tpHPR: repair time (>100 ms typical). One-time use (non-reversible).

sPPR (Soft PPR): Temporary row remap. Cleared on power cycle. Guard key required. tpSPPR: faster than hPPR.

MBIST PPR: Built-in self-test mode for identifying rows needing repair.

13.5 MPSM (Maximum Power Saving Mode)

- MR2 OP[1:0]=01B: enter MPSM
- All banks must be precharged before entry
- Lower power than self-refresh; used during extended idle
- Exit: issue DES then MRW to clear MPSM

13.6 Loopback Mode

- Test mode for characterization
- LBDQ, LBDQS pins: loopback output of DQ/DQS path
- MR enable + MPC command sequence to enter
- Output phase selectable (Table 368)
- Timing: tLBCK, tLBDQS per Table 369

13.7 Connectivity Test (CT) Mode

- MPC opcode 0001 0100
- DRAM drives all DQ pins with CMOS rail-to-rail output
- Used for PCB trace continuity testing
- RON_CT = 34 Ω (RONPu) / 34 Ω (RONPd)

13.8 3DS (3D Stacked) DDR5

- Stack heights: 2H, 4H, 8H (x4); 2H, 4H (x16)
- CID[3:0]: selects target die in stack
- Per-die addressing: same BG/bank/row structure per die
- tRFC3DS: larger tRFC for higher-density stacks (Table 283)
- tCCD_slr (same logical rank) / tCCD_dlr (different logical rank): separate timing paths
- 3DS speed bins: 3200–6400 with 2H/4H variants (Tables 476–490)

14 Self Refresh

- **Entry (SRE)**: all banks precharged; CS sequence per Table; CKE-equivalent implicit
- **During SR**: CK may stop; DRAM manages internal refresh; current = IDD6
- **Exit (SRX)**: stable clock required for tCKSRX before first command
- **tXS**: self-refresh exit to any non-MRS/REF command
- **tXSR**: SR exit to REF command (= tRFC + 10 ns)
- **IDD6** (SR current): device-density dependent, vendor-specified

15 MC Controller – DFI 5.2 Signal Mapping

15.1 DFI Command Encoding

JEDEC DDR5 commands → DFI 5.2 signals translation:

DDR5 Command	DFI Signal	Notes
ACT	dfi_address[17:0], dfi_bank[1:0], dfi_bg[2:0], dfi_act_n=L	2-cycle on DFI; freq_ratio determines nCK per DFI clk
RD / WR	dfi_address[10:0] (col), dfi_cas_n=L, dfi_we_n=L/H	dfi_rddata_en / dfi_wrdata_en issued ahead by RL/WL
PRE	dfi_ras_n=L, dfi_cas_n=H, dfi_we_n=L	AP: dfi_address[10]=H
REFab / REFsb	dfi_ras_n=L, dfi_cas_n=L, dfi_we_n=H	REFsb: dfi_bank[1:0] specifies bank
MRW / MRR	dfi_address encodes MRA[7:0] + OP[7:0]	2-cycle; Init FSM drives directly
NOP / DES	dfi_cs_n=H or all dfi_*=deassert	–

15.2 Key DFI 5.2 Signals

DFI Signal	Function
dfi_address[17:0]	Row/column/mode register address
dfi_bank[1:0]	Bank address
dfi_bg[2:0]	Bank group (DDR4/5)
dfi_act_n	ACT command indicator (DDR4/5)
dfi_cs_n[ranks-1:0]	Chip select per rank
dfi_wrdata[n:0]	Write data bus
dfi_wrdata_en	Write data enable; issued WL cycles before data
dfi_wrdata_mask[n/8:0]	DM mask bits
dfi_rddata[n:0]	Read data from PHY
dfi_rddata_valid	PHY asserts when rddata valid
dfi_rddata_en	Controller assertion RL cycles before CAS
dfi_ctrlupd_req/ack	Controller-initiated PHY update handshake
dfi_phyupd_req/ack	PHY-initiated update request
dfi_init_start/complete	Init sequence handshake
dfi_freq_ratio	1:1, 1:2, 1:4 ratio of MC clk to DRAM clk
dfi_dram_clk_enable	CK enable to PHY
dfi_t_rddata_en	Latency param: rddata_en to valid in PHY clocks
dfi_t_phy_wrlat	Latency param: wrdata_en to DFI write data

Source: JEDEC Standard JESD79-5 (July 2020), 496 pages. All table/section numbers reference the original JEDEC document. Timing values are representative; always validate against device datasheet and speed bin. **DFI**: MIPI DFI Specification v5.20.